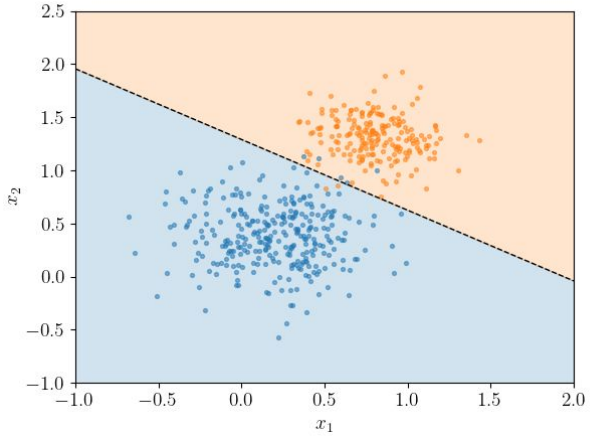
Lecture 5: Deep learning basics

Ran Zhang
11/02/2026

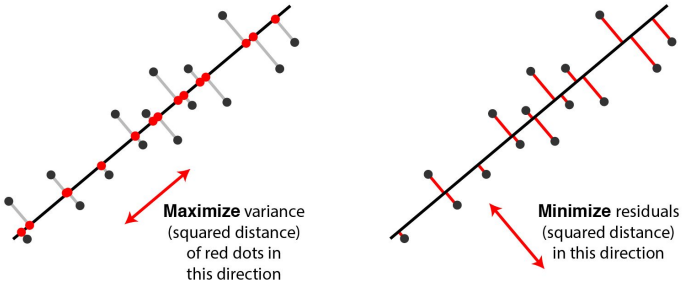
Recap

Logistic regression



Learns a linear decision boundary

PCA

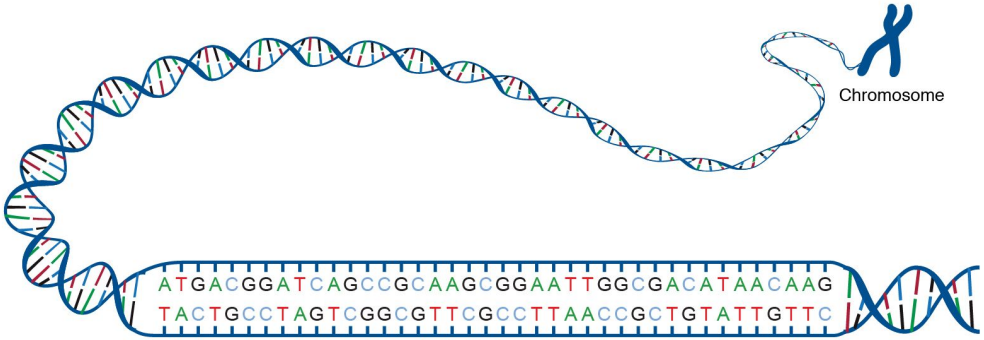


Assumes that a linear projection is sufficient for dimensionality reduction

Biological data often contain complex and nonlinear relationships

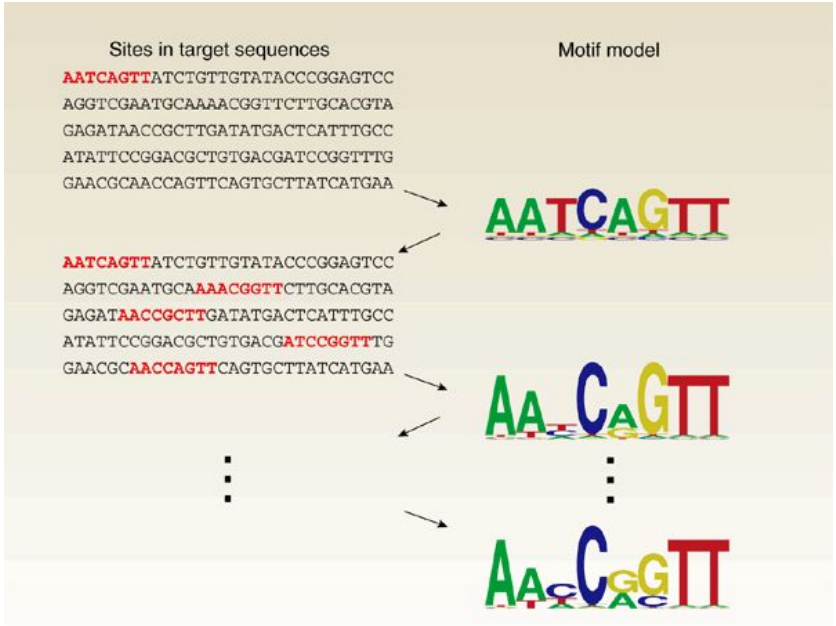


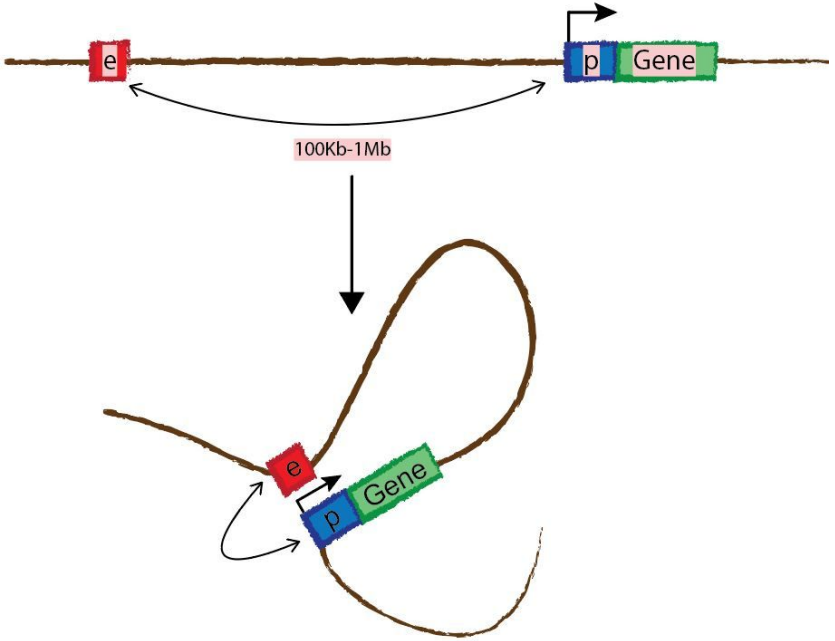
Medical images



DNA sequences

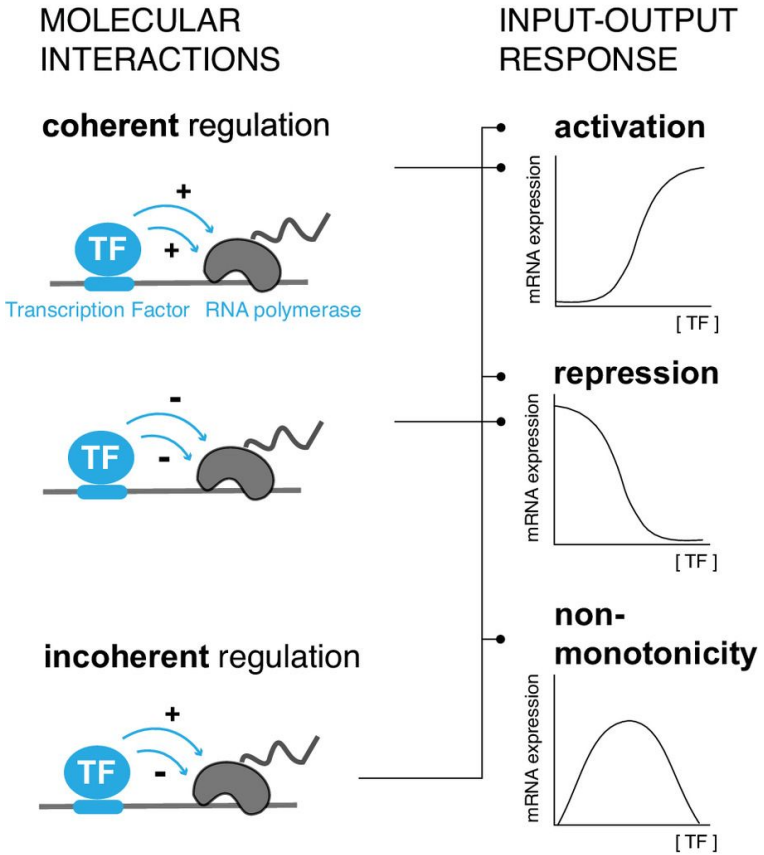
We need nonlinear methods to capture complex relationships among features





Patrik D'haeseleer. 2006
https://www.fmi.ch/news-events/articles/news.html?news=542

We need nonlinear methods to capture complex relationships among features



Martinez-Corral et al. 2025

Today's class

- Neural network basics:
  - Nodes and hidden layers
  - Multilayer perceptron
  - Training a neural network
- Example applications

Our brains make predictions for complex tasks all the time







Danger!

Safe!

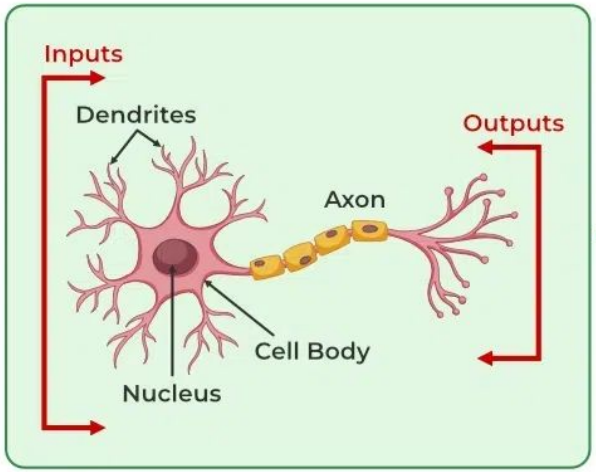
Wikipedia &
https://www.freepik.com/free-photos-vectors/simple-cartoon-human

Neurons in our brain converts input signals to a binary prediction





Neuron

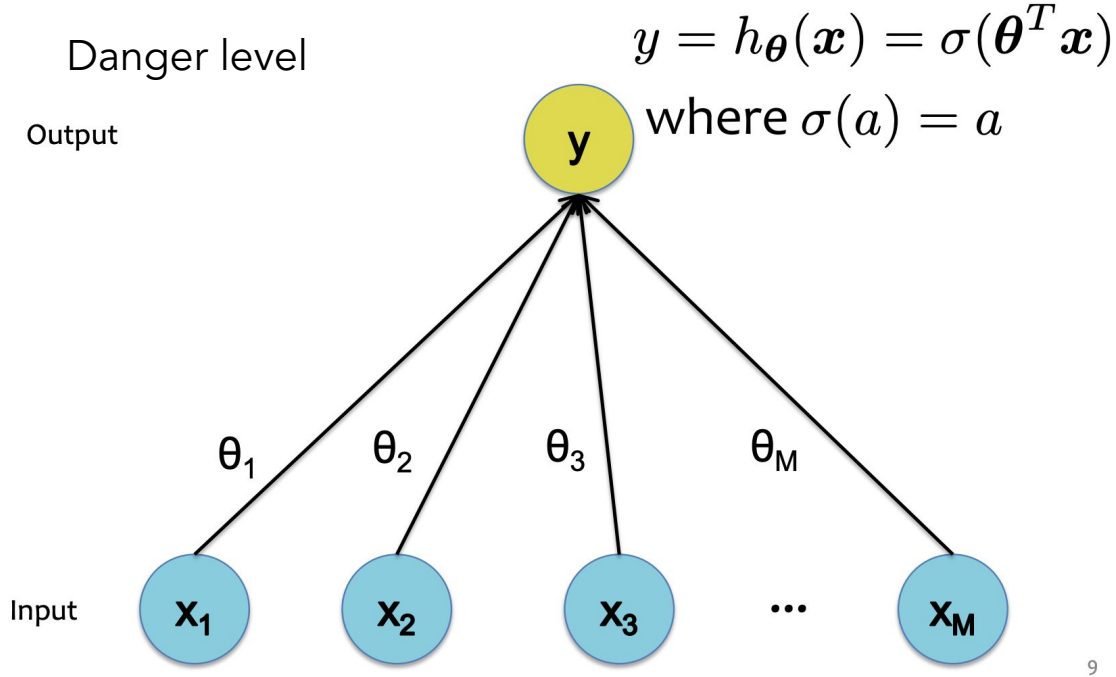


Danger!

Safe!

https://www.freepik.com/free-photos-vectors/simple-cartoon-human

Linear regression output is the weighted sum of input signals





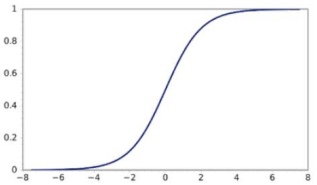


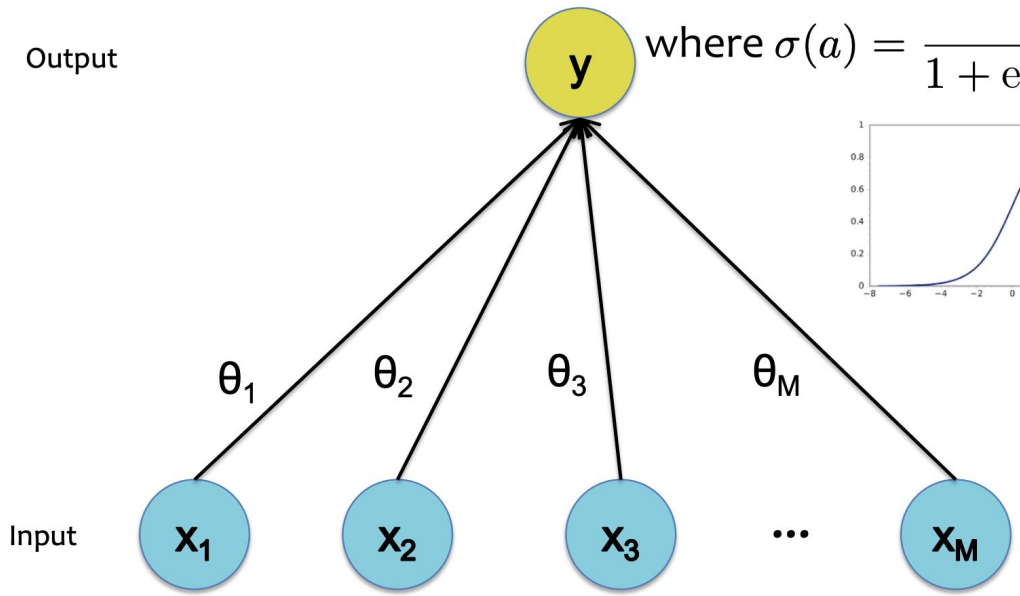
Adapted from https://www.cs.cmu.edu/~mgormley/courses/10601-f19/slides/lecture12-nn.pdf

Logistic regression adds a sigmoid transformation to the output

$$y = h_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sigma(\boldsymbol{\theta}^T \boldsymbol{x})$$

where $\sigma(a) = \dfrac{1}{1 + \exp(-a)}$





This can also be viewed as a perceptron in neural network

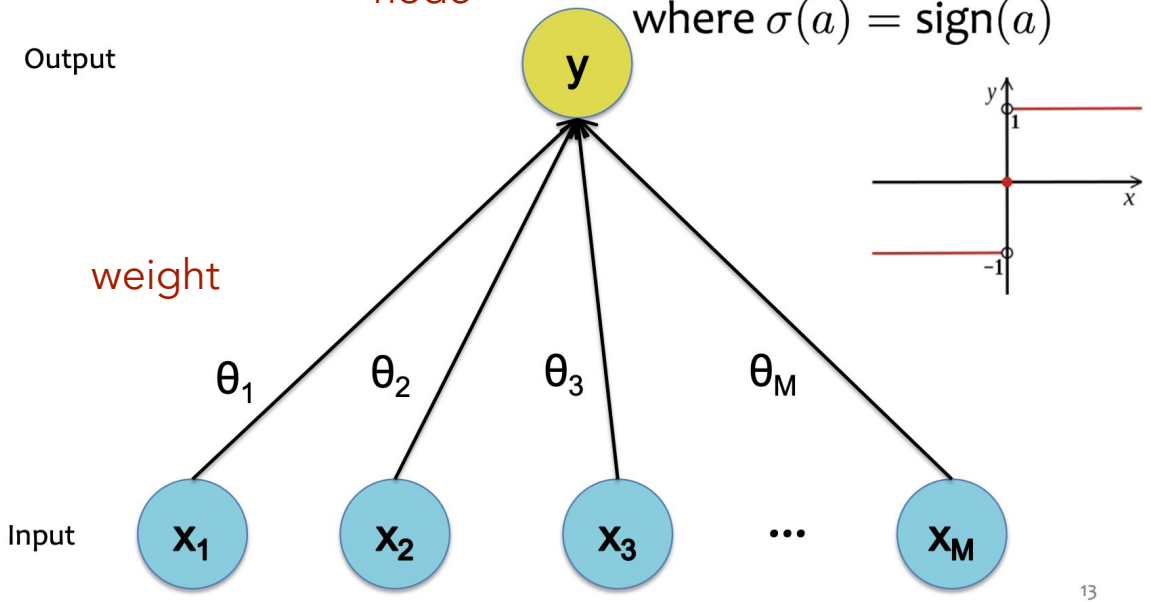
Adapted from https://www.cs.cmu.edu/~mgormley/courses/10601-f19/slides/lecture12-nn.pdf

A perceptron (neuron) is a generalized form of logistic regression

Perceptron

$$y = h_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sigma(\boldsymbol{\theta}^T \boldsymbol{x})$$

where $\sigma(a) = \text{sign}(a)$



Activation function

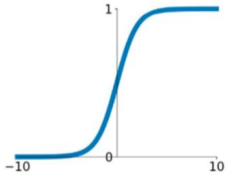
13

Adapted from https://www.cs.cmu.edu/~mgormley/courses/10601-f19/slides/lecture12-nn.pdf

There are many types of activation functions to choose from

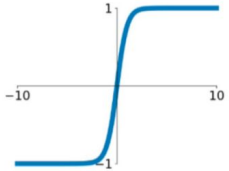
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



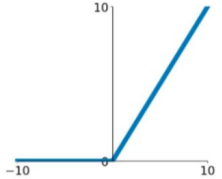
tanh

$$\tanh(x)$$



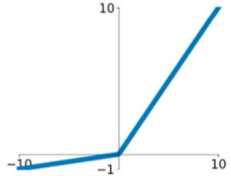
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

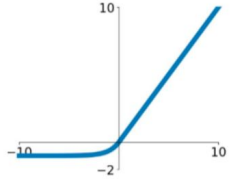


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$

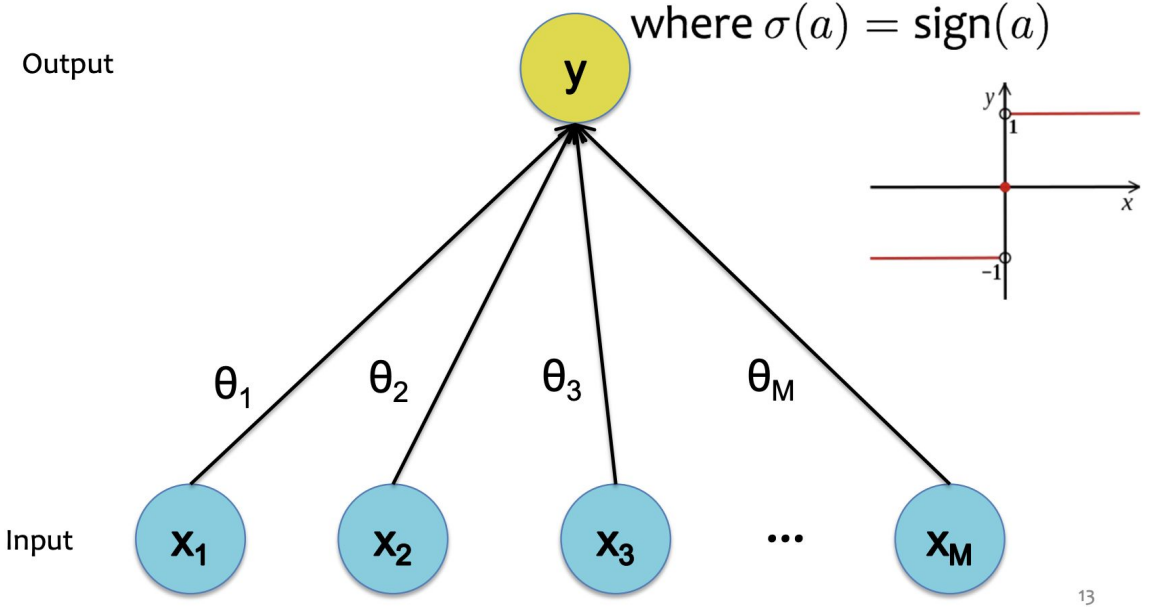


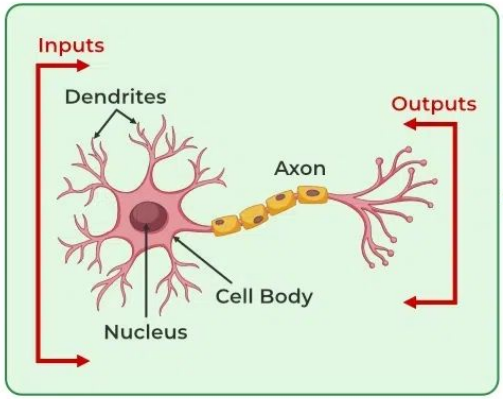
https://storage1.ucsd.edu/slides/CSE152/L6_NeuralNetwork.html#/--ActivationFunctions

A perceptron is the basic building block of a neural network - similar to a neuron in the brain

$$y = h_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sigma(\boldsymbol{\theta}^T \boldsymbol{x})$$

where $\sigma(a) = \text{sign}(a)$





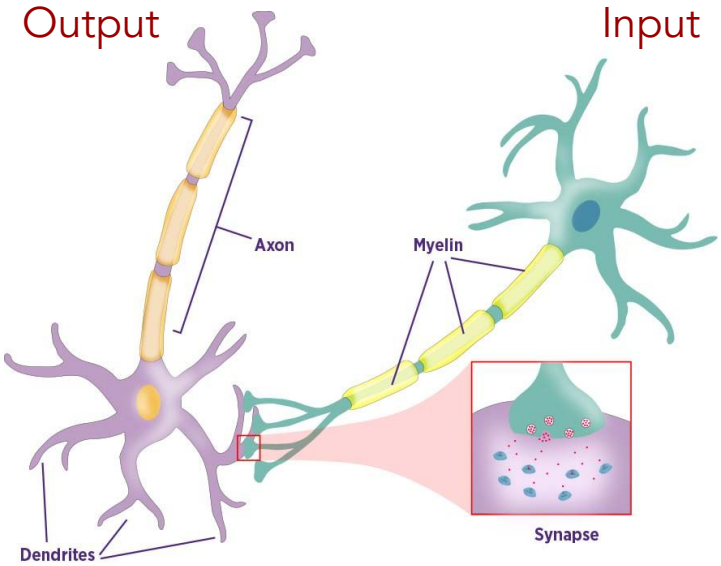
13

The brain works by signal transduction among neurons

In the brain, a neuron:

- receives signals from many other neurons
- combines and integrates those signals
- sends output signals to other neurons

There are 86 billion neurons in human brain!

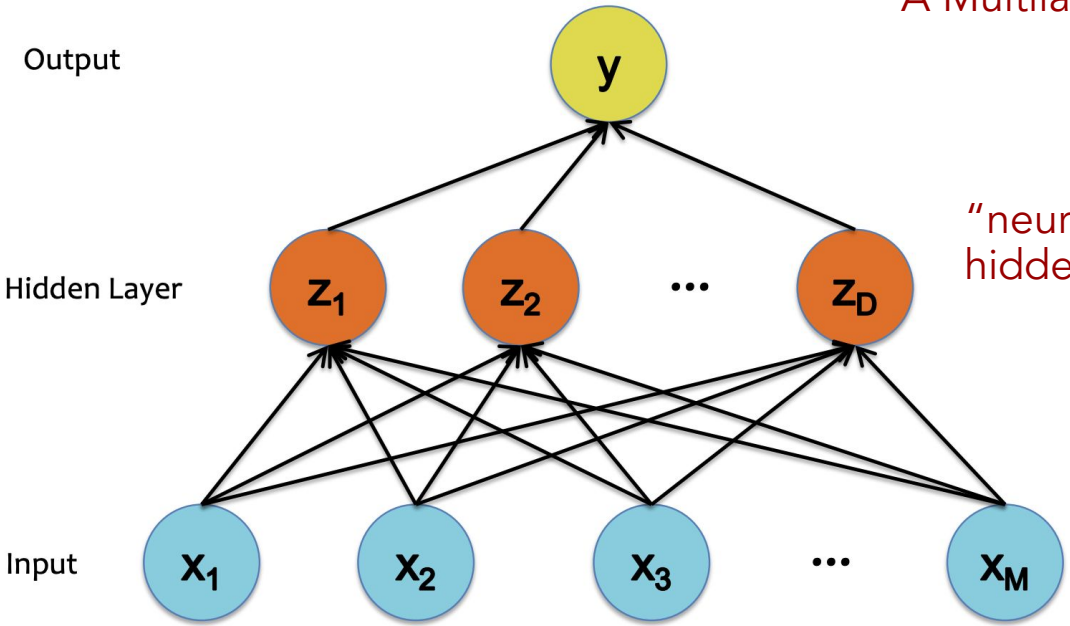




Queensland Brain Institute

A neural network also consists of multiple layers of perceptrons stacked together

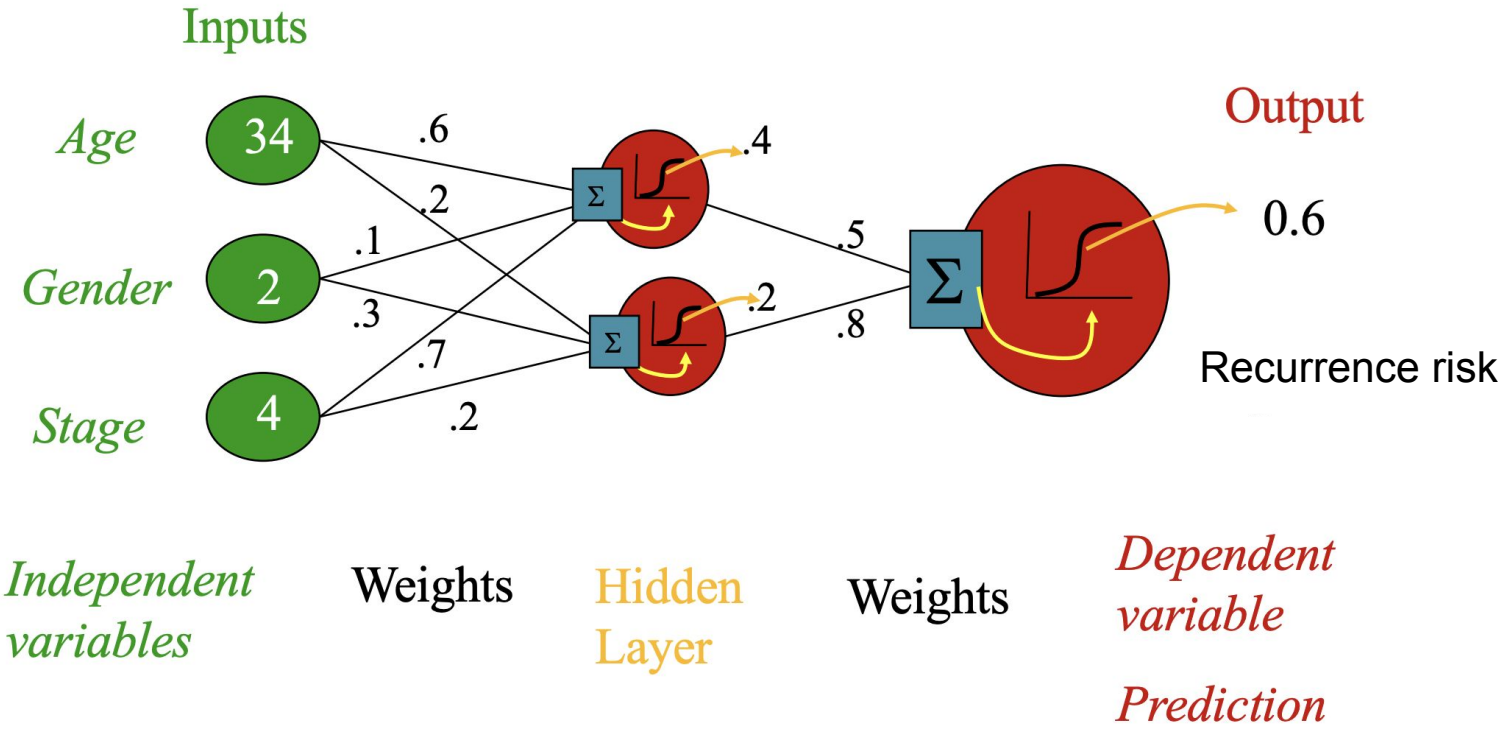
A Multilayer Perceptron (MLP)



"neuron" – hidden node

Adapted from https://www.cs.cmu.edu/~mgormley/courses/10601-f19/slides/lecture12-nn.pdf

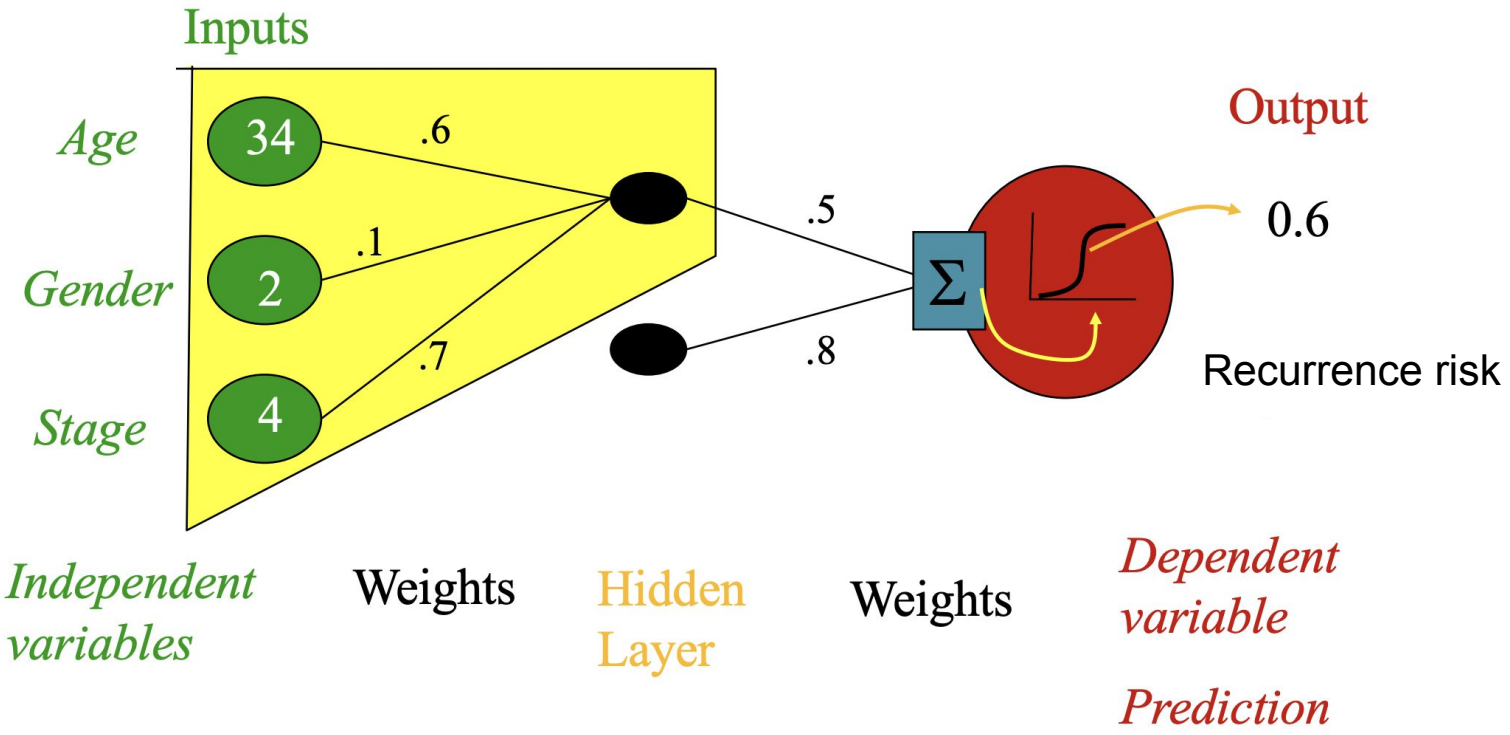
Example: how to predict cancer recurrence risk based on age, gender and cancer stage?



Adapted from https://www.cs.cmu.edu/~mgormley/courses/10601-f19/slides/lecture12-nn.pdf

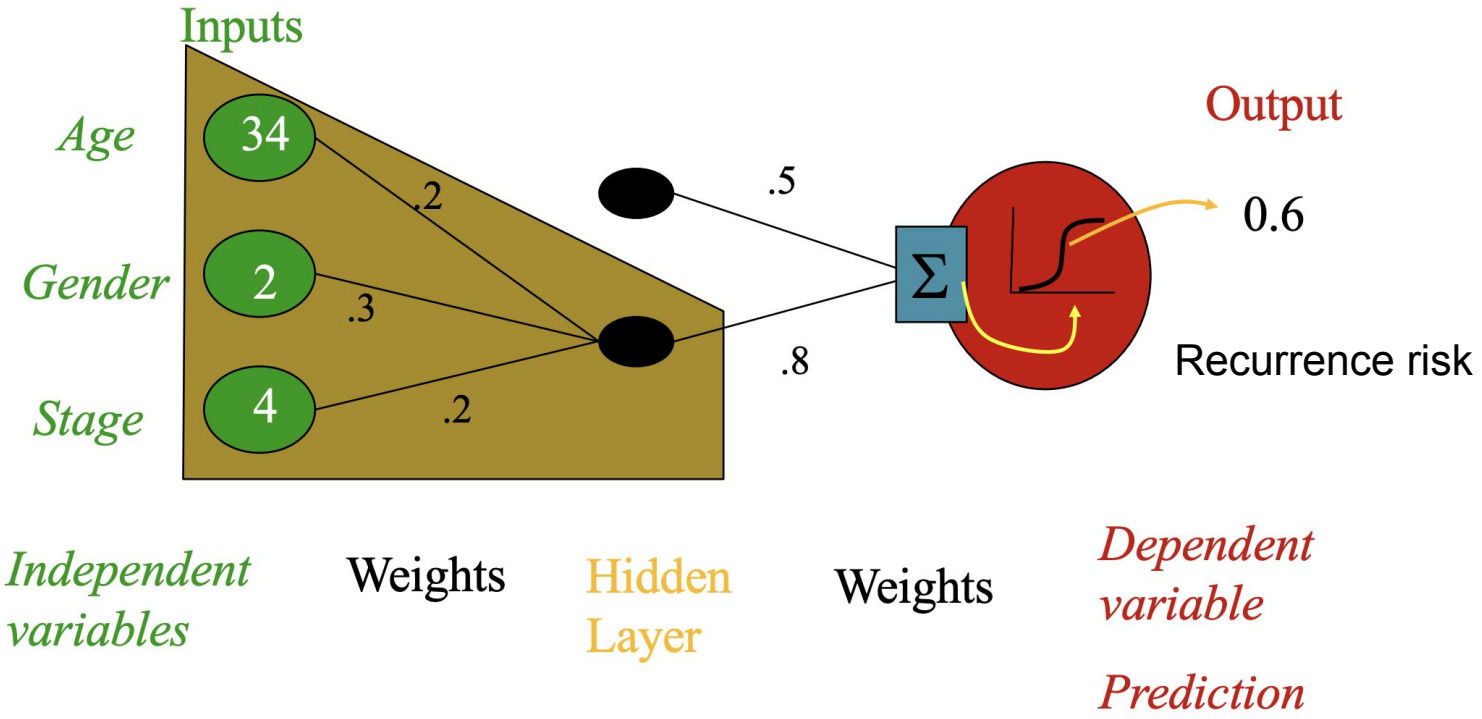
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A neural network nonlinearly combines input signals to predict the output



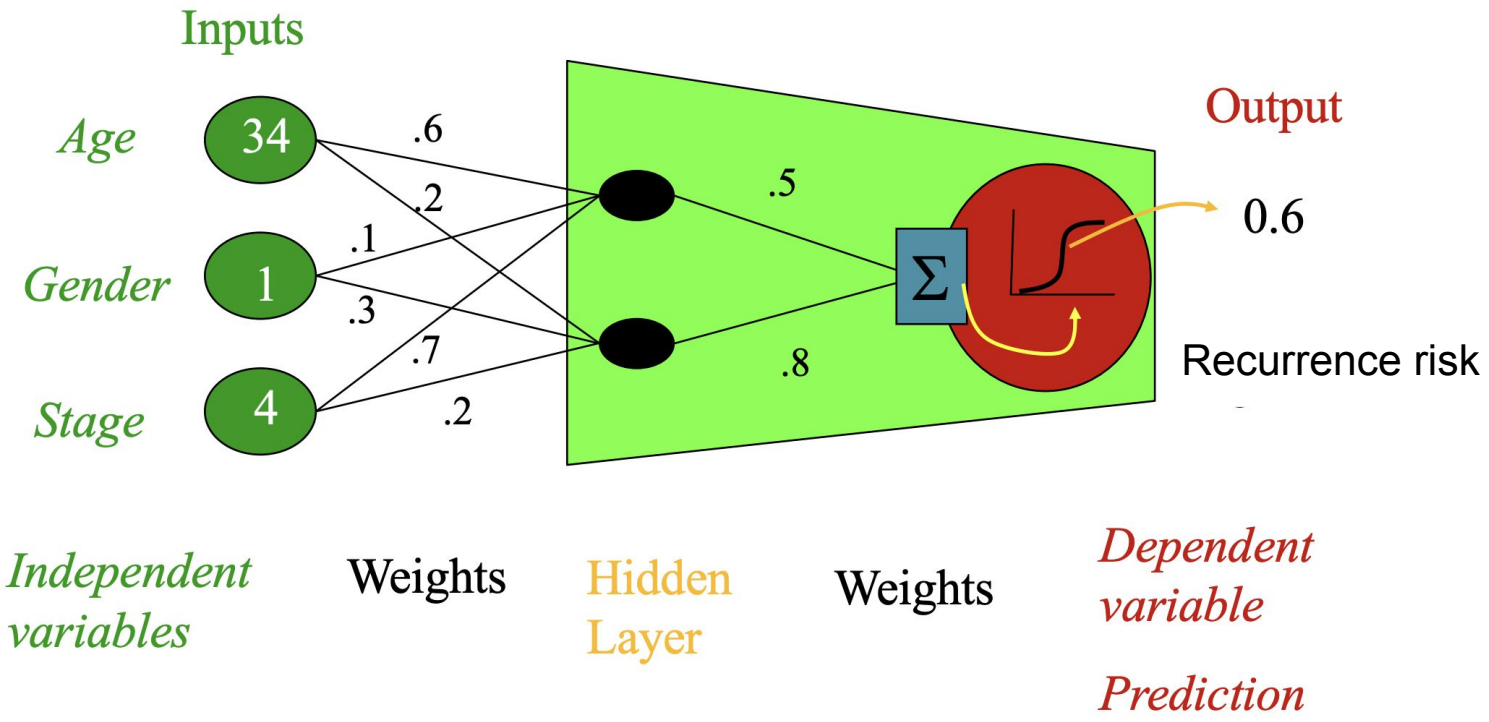
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A neural network nonlinearly combines input signals to predict the output



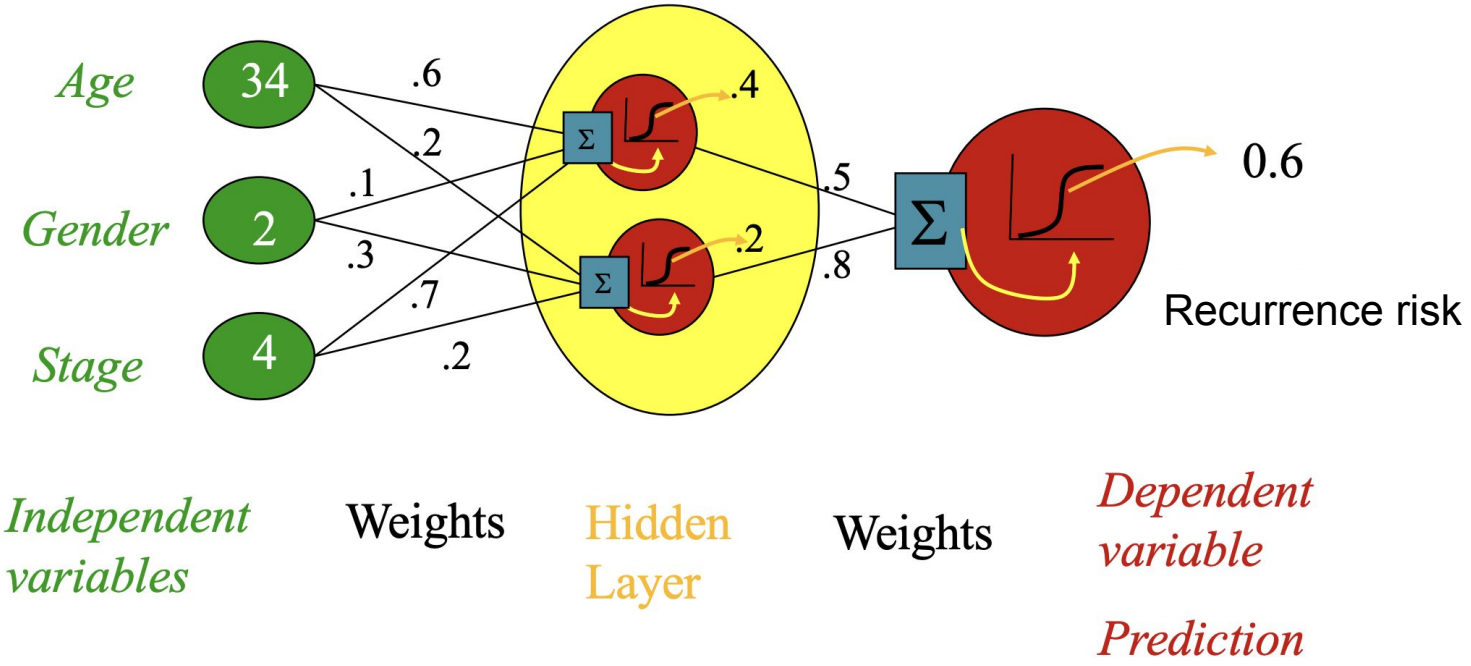
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A neural network nonlinearly combines input signals to predict the output



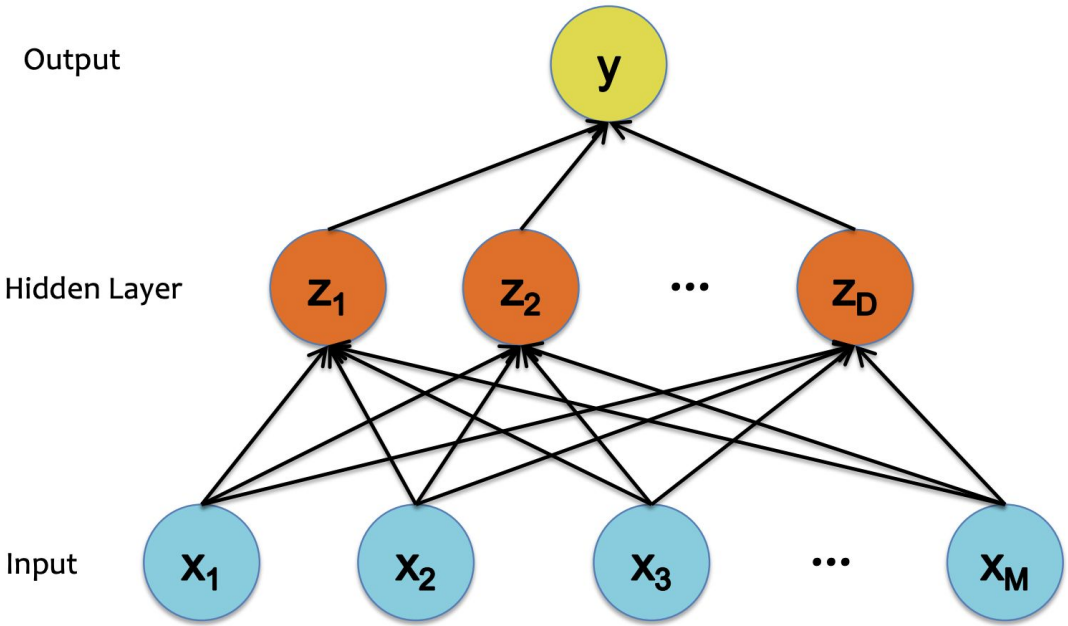
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A neural network nonlinearly combines input signals to predict the output

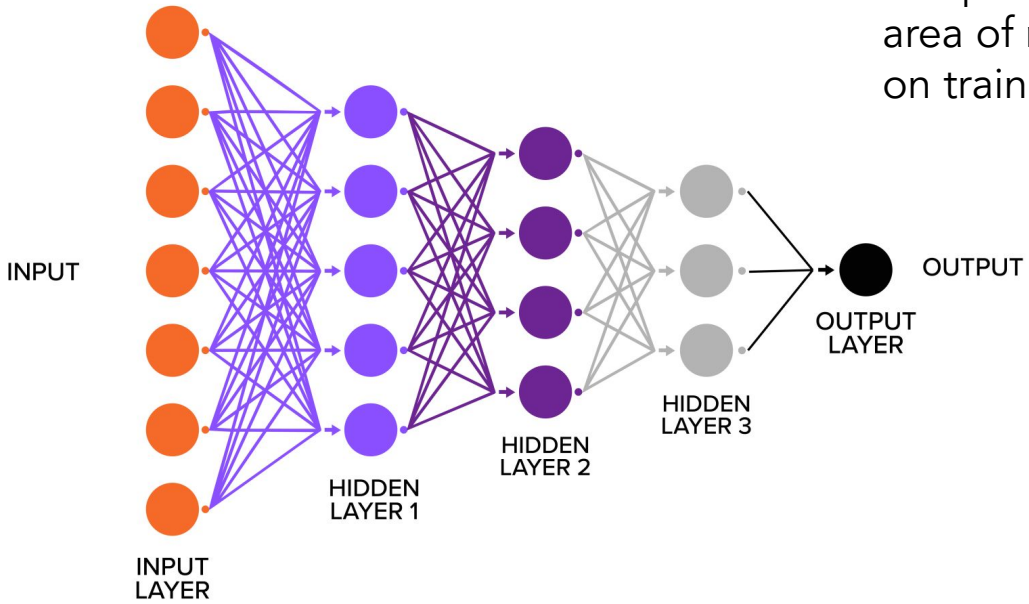


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Question: what if no activation function is applied to each node?



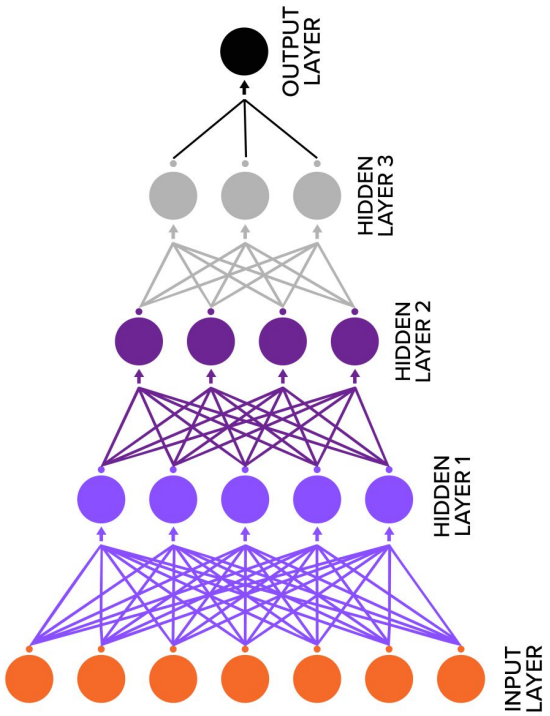
A deep neural network consists of many hidden layers and can learn complex patterns in data

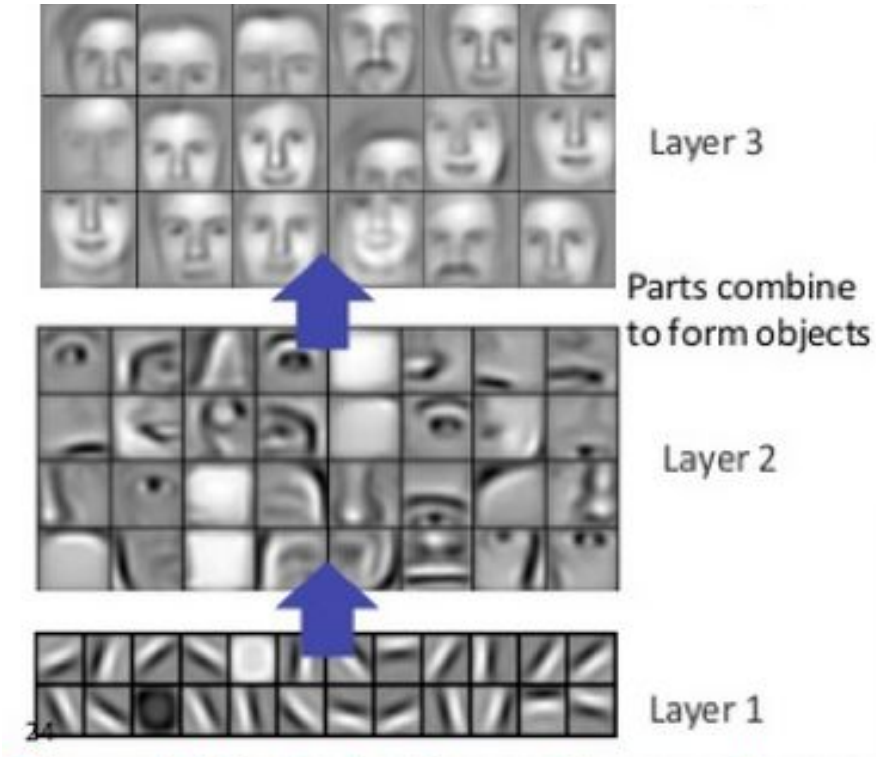


Deep learning refers to the broader area of machine learning that focuses on training deep neural networks.

https://www.smartboost.com/blog/deep-learning-vs-neural-network-whats-the-difference/

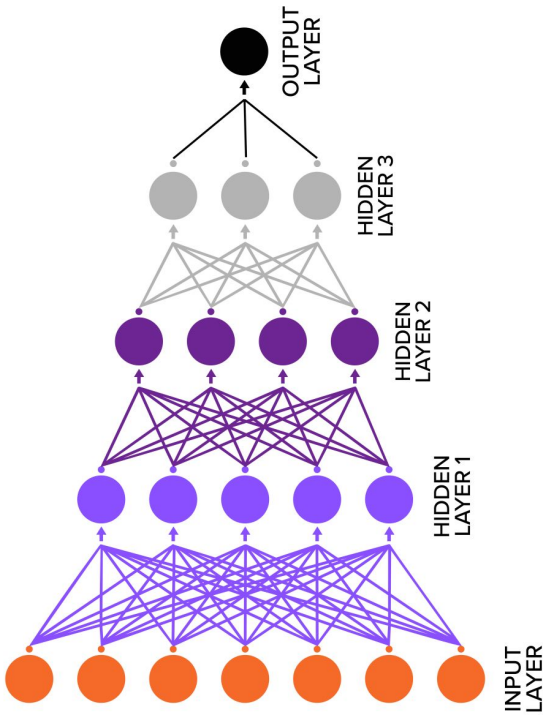
The further into the neural net, the more complex the features nodes can recognize





https://wiki.pathmind.com/neural-network

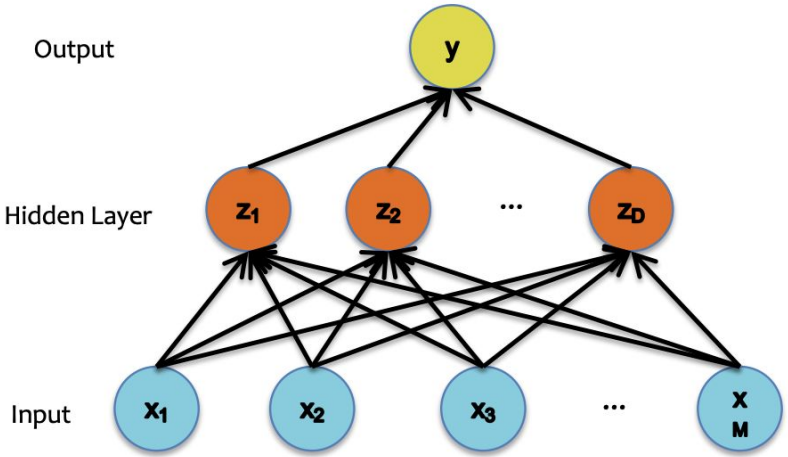
Neural networks can handle very complex patterns

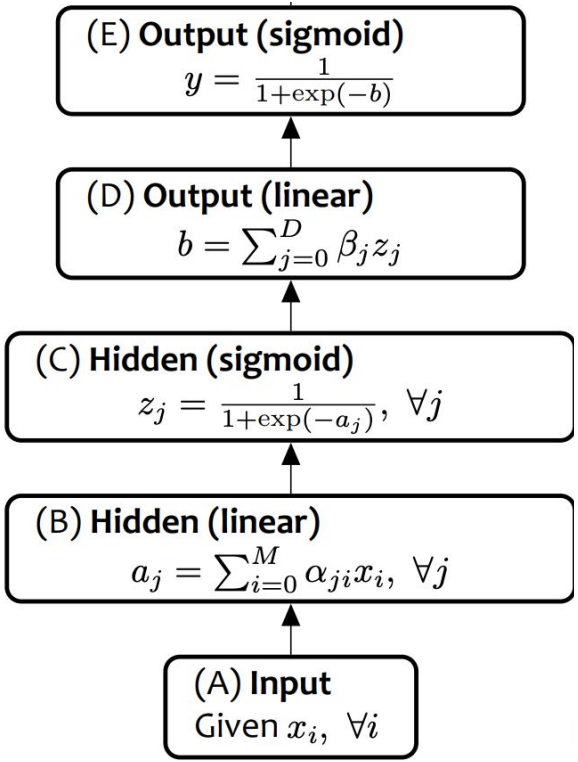


- Each layer can be associated with different activation functions - learns complex relationships
- Handles different types of input
- Way more flexible than other non-linear methods such as SVM and random forests

https://www.smartboost.com/blog/deep-learning-vs-neural-network-whats-the-difference/

How to train a neural network?





https://www.cs.cmu.edu/~mgormley/courses/10601-f19/slides/lecture12-nn.pdf

Recap: loss function for machine learning tasks

Logistic regression $$L = -\sum_{i=1}^{n} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Decision tree $$G = 1 - \sum_{k=1}^{K} p_k^2$$

Support vector machine $$L = \sum_{i=1}^{n} \max(0, 1 - y_i(w^T x_i + b))$$

https://www.cs.cmu.edu/~mgormley/courses/10601-f19/slides/lecture12-nn.pdf

Neural networks can be trained using a variety of loss functions

Regression (Mean squared error, similar to linear regression)

$$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Classification
(Binary cross entropy, similar to logistic regression)

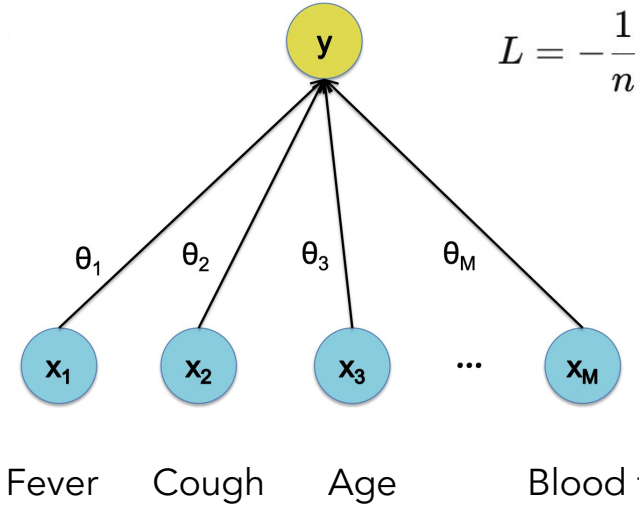
$$L = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$$

https://www.cs.cmu.edu/~mgormley/courses/10601-f19/slides/lecture12-nn.pdf

How to train a neural network to predict whether a person have flu?

Does the person have flu? Yes (1) No (0)





Loss function:

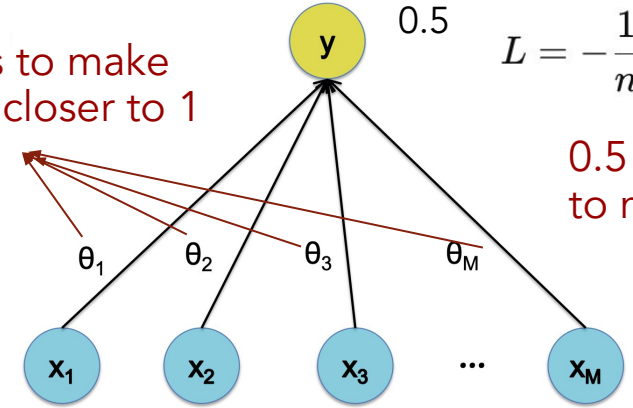
$$L = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$$

When the doctor makes errors, the doctor adjusts the neural network weights

Does the person have flu? Yes (1) No (0)



Adjust weights to make the prediction closer to 1



Fever 1 | Cough 1 | Age 20 | ... | Blood test 1

Loss function:

$$L = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$$

0.5 is not close enough to 1, how to minimize the loss function?

Has flu

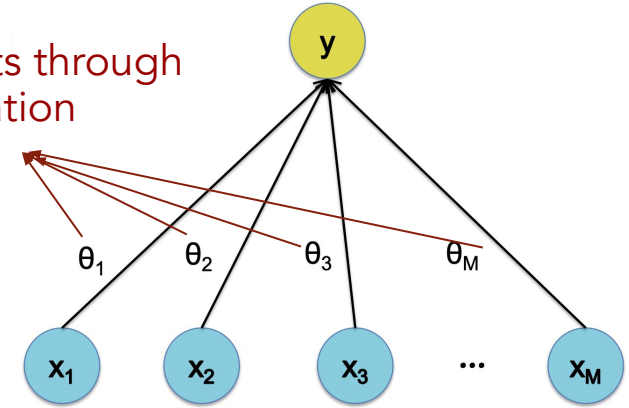


The model keeps getting updated as the doctor sees more and more patients to improve prediction accuracy

Does the person have flu? Yes (1) No (0)



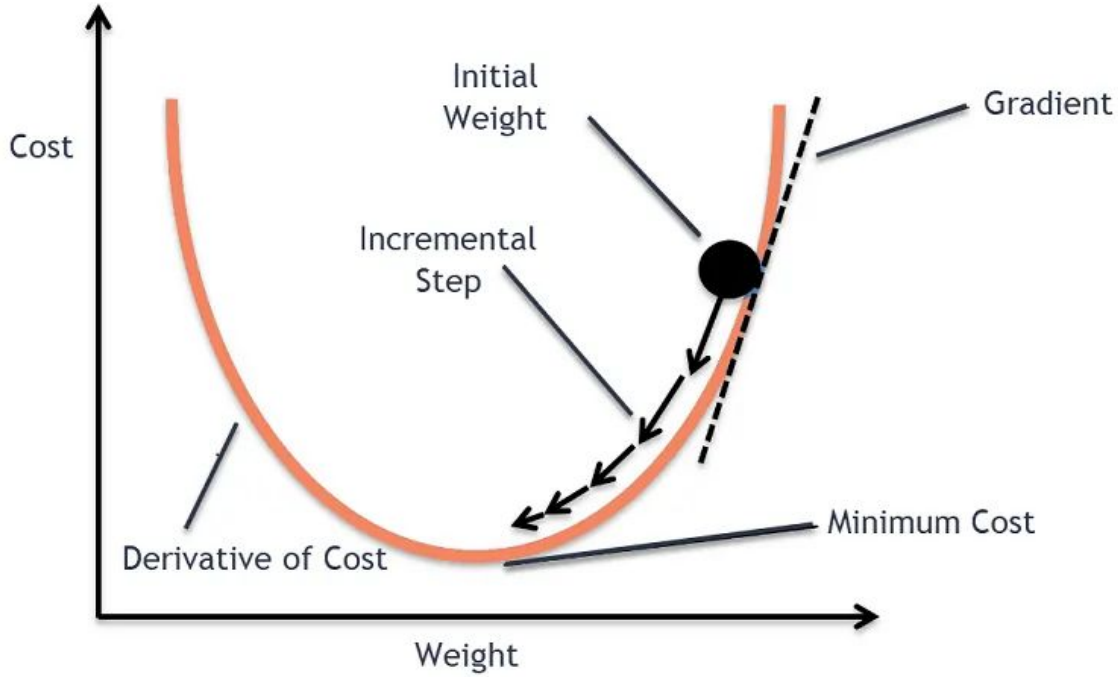
Adjust weights through back propagation



Fever	Cough	Age	Blood test
1	1	20	1
…	…	…	…



Backpropagation computes gradients that guide the ball downhill toward the lowest cost in small steps



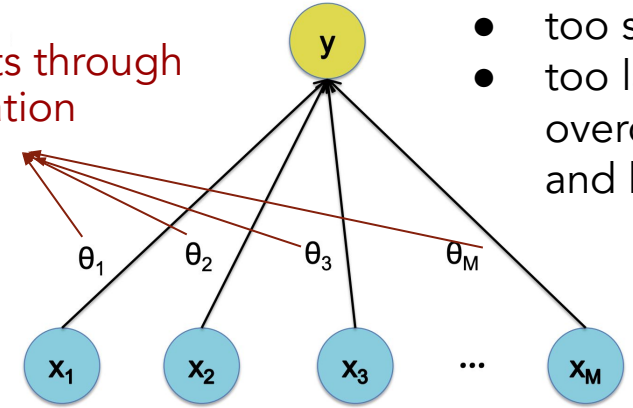
https://www.analyticsvidhya.com/blog/2020/10/how-does-the-gradient-descent-algorithm-work-in-machine-learning/

How fast should the doctor adjust the weight?

Does the person have flu? Yes (1) No (0)



Adjust weights through back propagation



Fever 1 …

Cough 1 …

Age 20 …

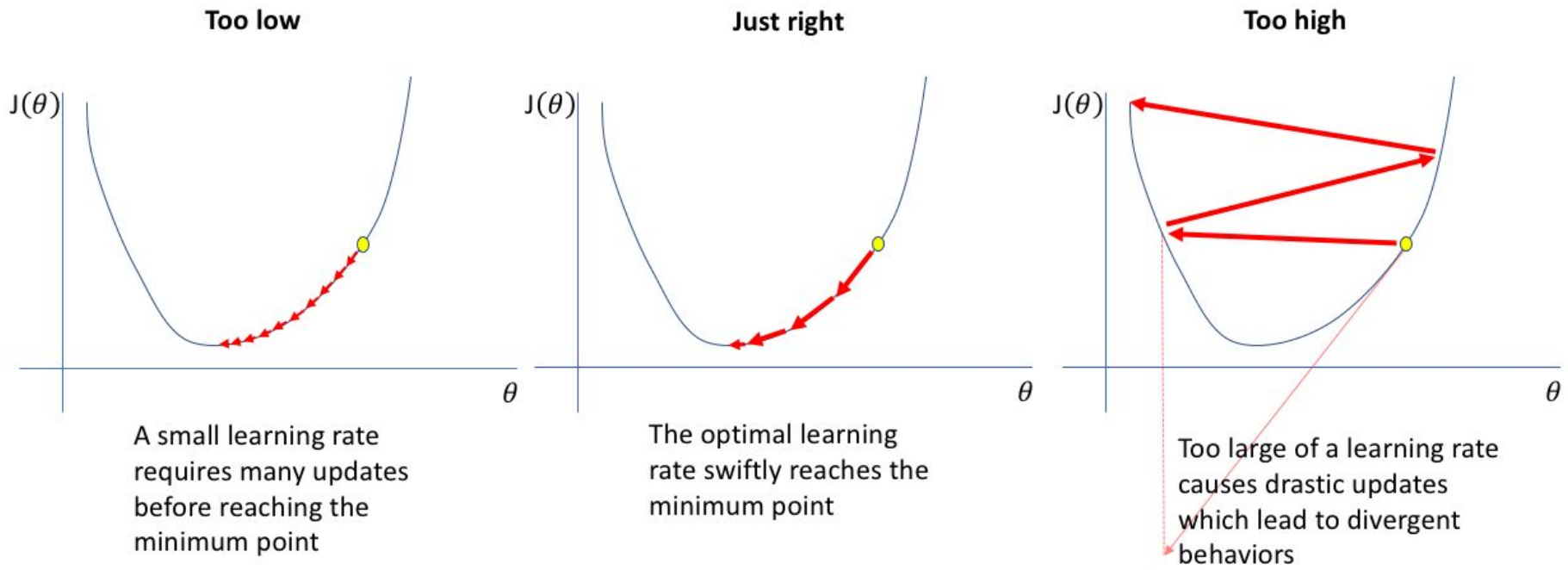
Blood test 1 …

- too small → learning is very slow
- too large → the model can get overcorrected by one or two outliers and becomes unstable

"Learning rate"



A model may not reach optimal solution with very high and very low learning rates

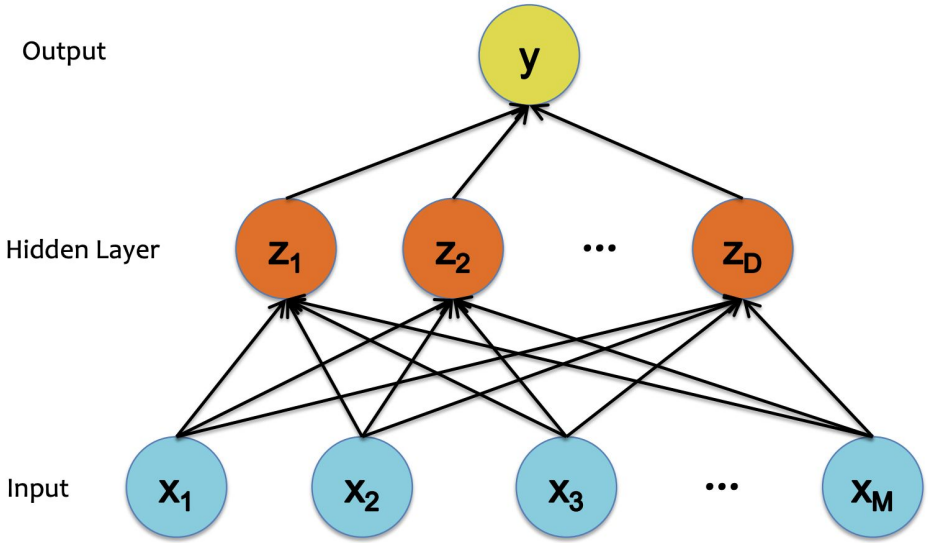


https://www.jeremyjordan.me/nn-learning-rate/

There are several key hyperparameters to decide for each neural network

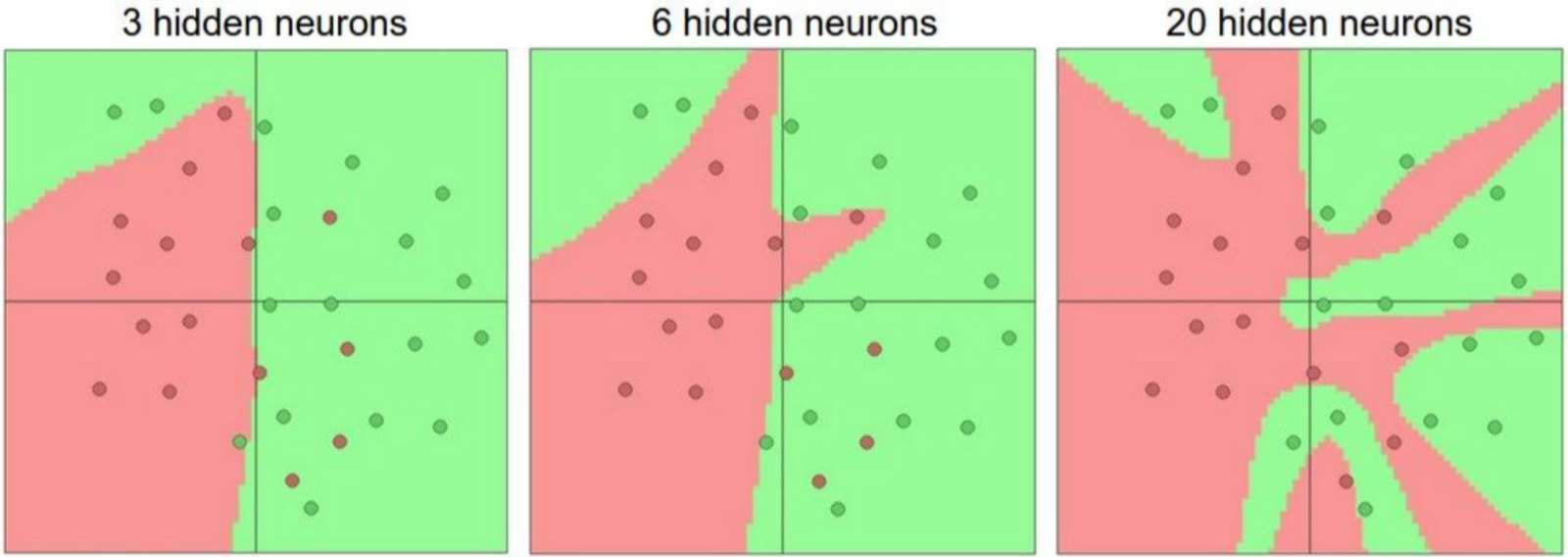
Hyperparameters:

- Learning rate
- Number of hidden layers
- Number of nodes in each hidden layer



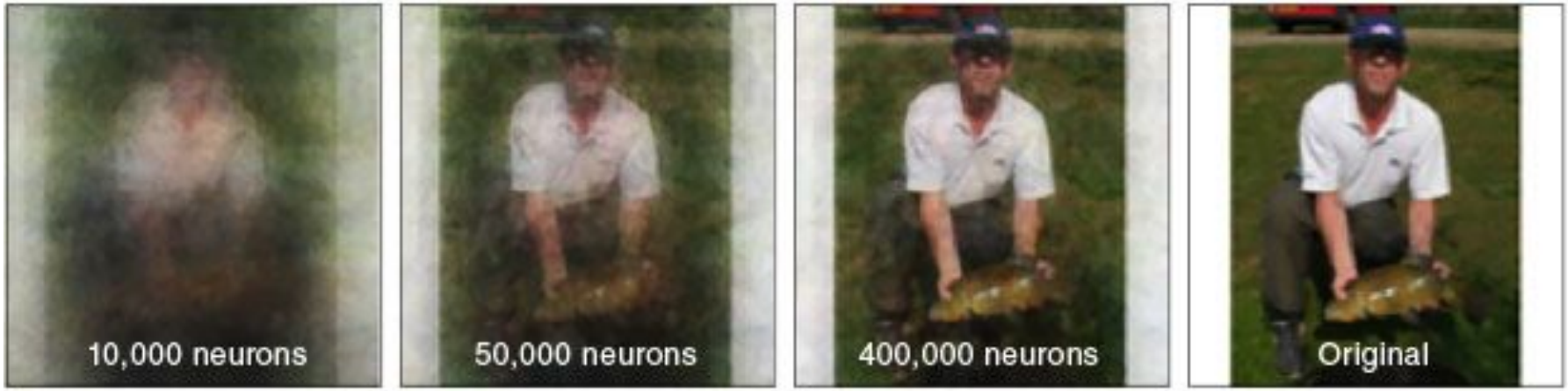
- too small → model is not able to learn complex patterns
- too large → the model becomes unstable and overfits

Decision boundary gets more and more complex as the number of hidden neurons increases



https://storage1.ucsd.edu/slides/CSE152/L6_NeuralNetwork.html#/--DecisionBoundariesofNeuralNetworks_2

Decision boundary gets more and more complex as the number of hidden neurons increases



https://str.llnl.gov/past-issues/june-2016/deep-neural-networks-bring-patterns-focus

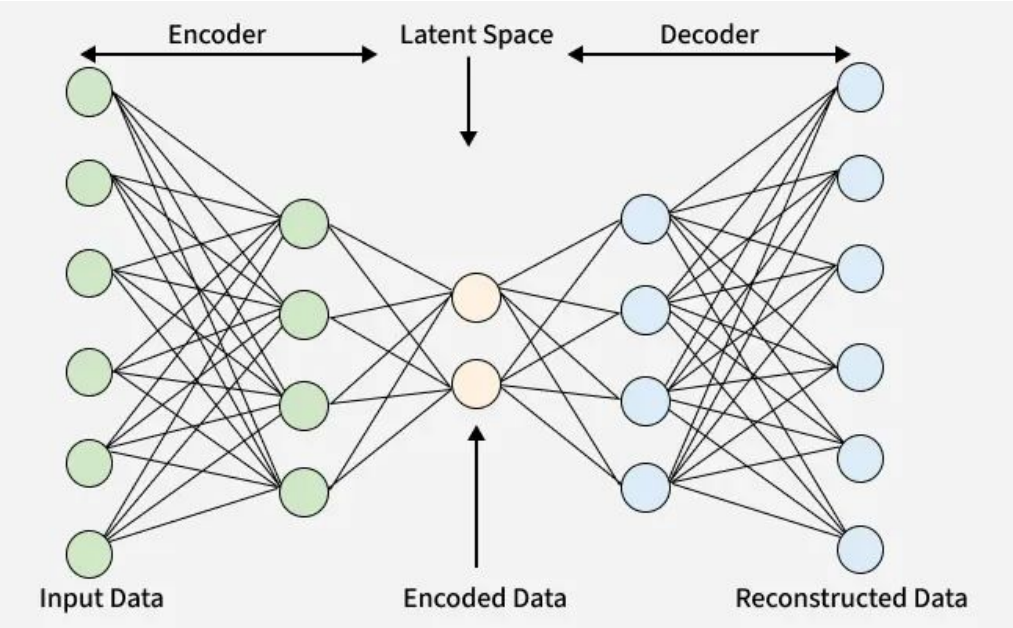
Neural network excels at handling complex cases with a lot of observations

Model	Best When
Neural Network	Large dataset, complex patterns
Linear regression	Linear relationships, interpretability is important
SVM	Small sample, high dimension
Random Forest	Noisy small-to-medium data with mixed feature types

Example use cases: unsupervised cell clustering

- Single-cell RNA-seq: each cell has ~20k genes measured, and the majority of the features are sparse due to technical limitations.
- We want to find cell clusters and annotate cell types
- How to project sparse and noisy scRNA-seq measurements to low-dimensional space?

Autoencoders convert high-dimensional measurements to low-dimensional representations while preserving important information

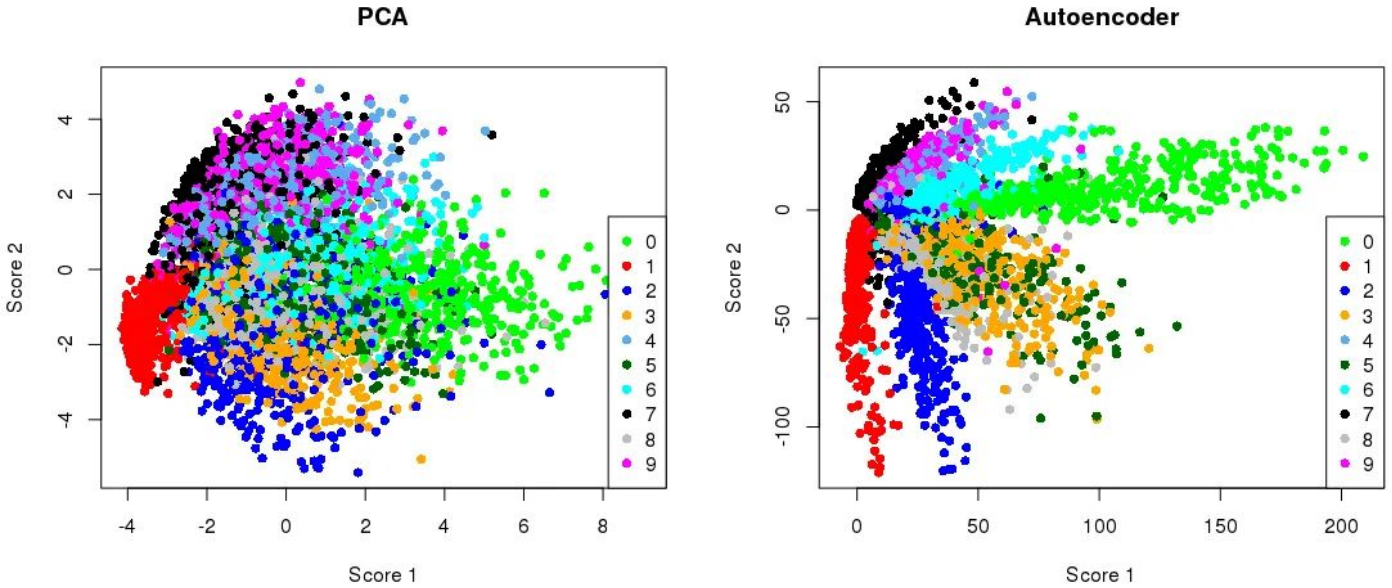


- Loss = MSE (input, output)
- The latent representation contains condensed information for downstream clustering / prediction tasks

https://www.geeksforgeeks.org/numpy/types-of-autoencoders/

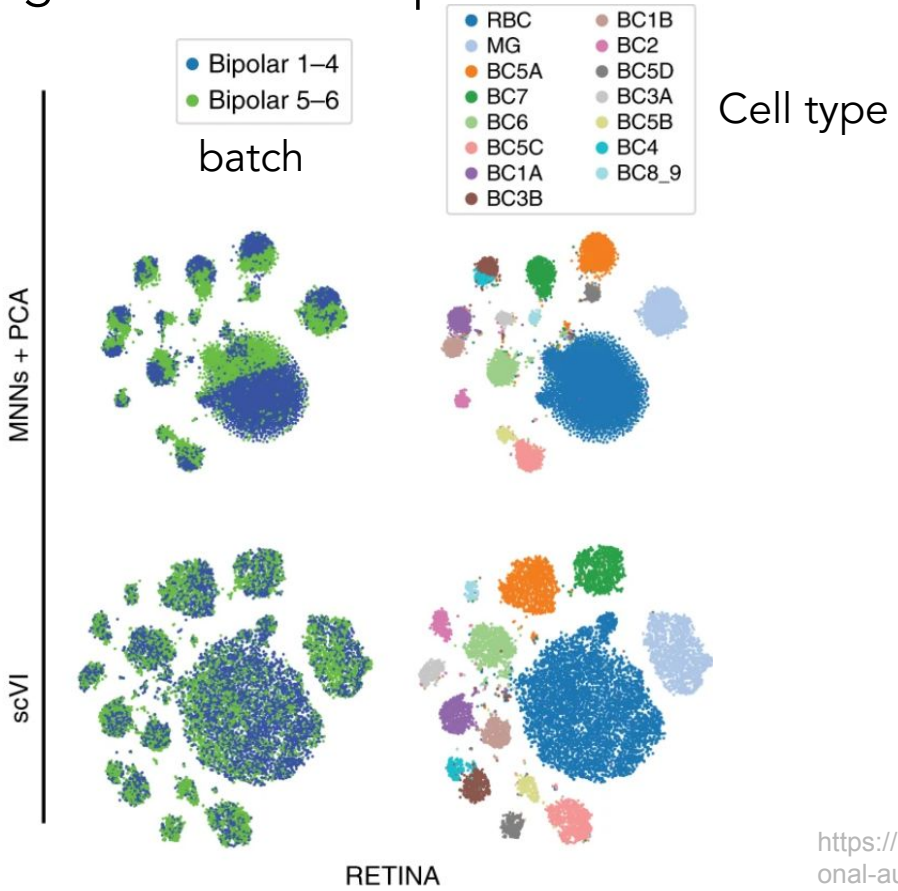
Autoencoders can be viewed as a nonlinear generalization of PCA that can learn more flexible representations

- PCA performs linear dimensionality reduction
- Autoencoders learn nonlinear low-dimensional representations



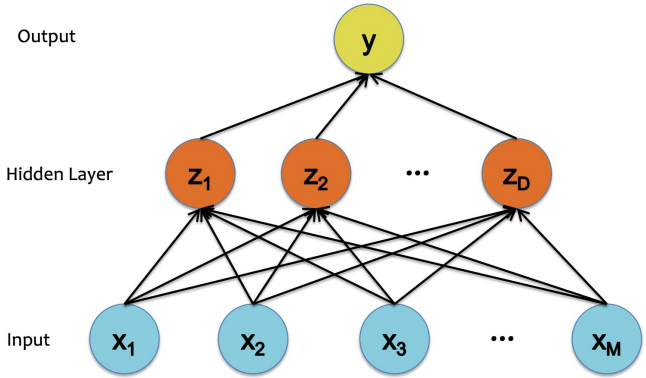
https://medium.com/@Ravitejakandimalla/variational-autoencoder-vs-pca-pytorch-3469e7ad12a1

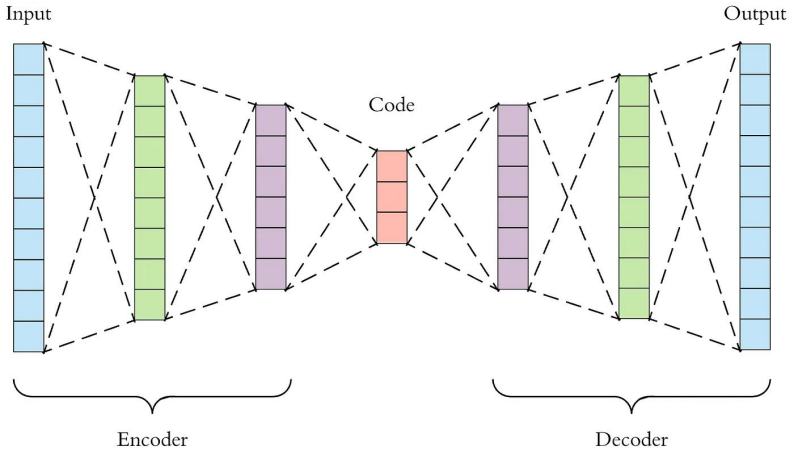
Autoencoders (scVI) enables better cell type separation and batch correction for single-cell RNA-seq data



https://medium.com/@Ravitejakandimalla/variational-autoencoder-vs-pca-pytorch-3469e7ad12a1

So far, we are formatting input as independent nodes





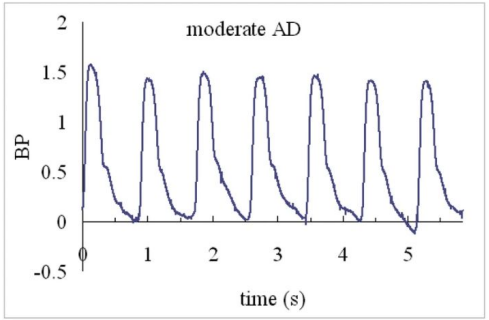
- If we switch features x_1 and x_3, does it make any difference?

In biological and health science studies, measurements usually are not independent from each other

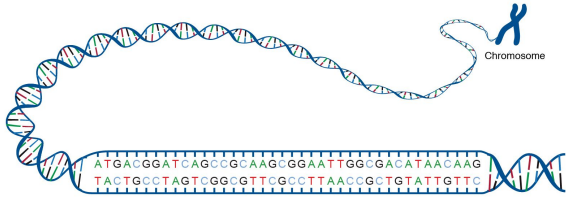
imaging



EHR



DNA sequence



- Motivation: how to use contextual information to help with our model and prediction?

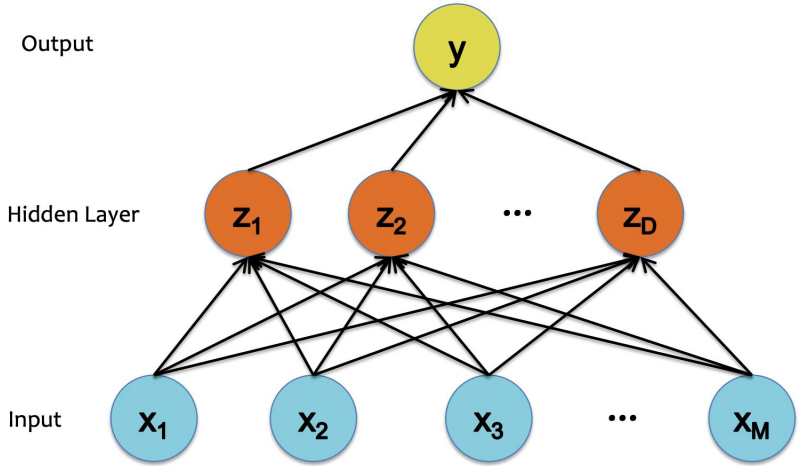
Multilayer perceptron need to learn a lot of weight parameters, which can lead to overfitting

imaging



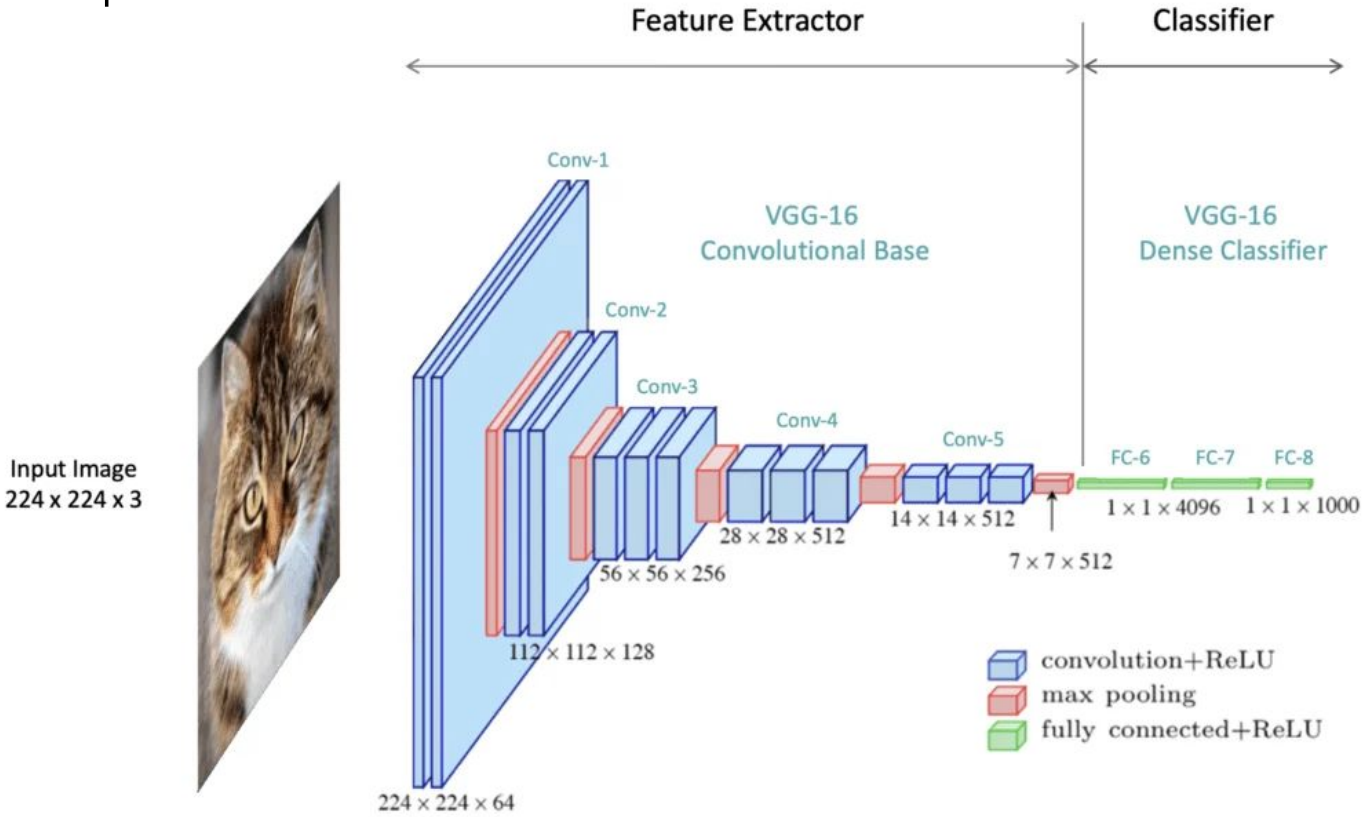
~ 1000 hidden nodes

~ 2 million pixels

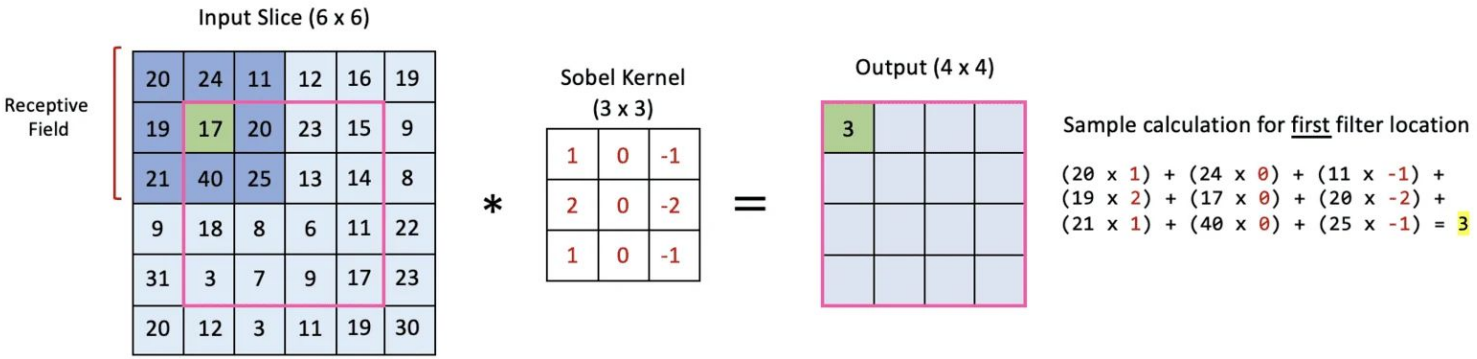


- Number of weights to learn for the first hidden layer: 2 million x 1000 = 2 billion
- The architecture ignores local patterns and treat each pixel independently

Convolutional neural networks (CNN) focus on step-wise aggregation of local patterns

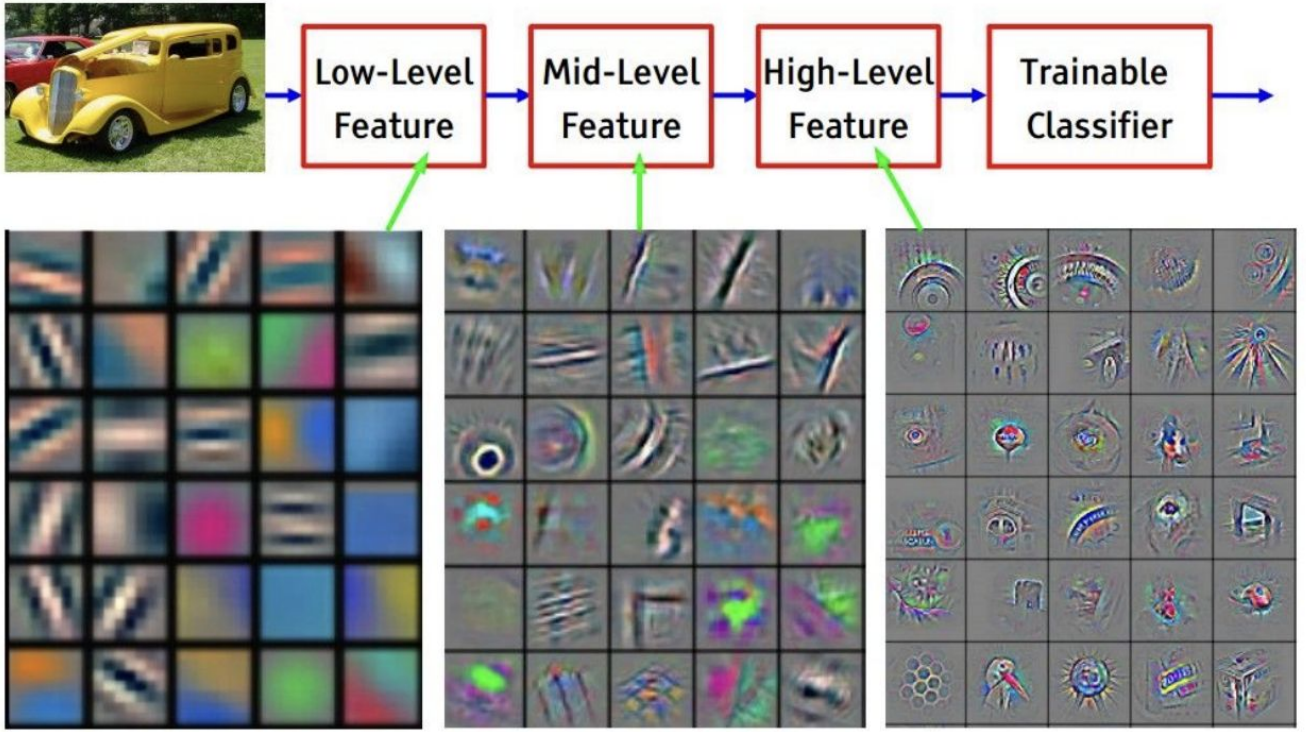


At each step, local patterns are aggregated into higher-level patterns



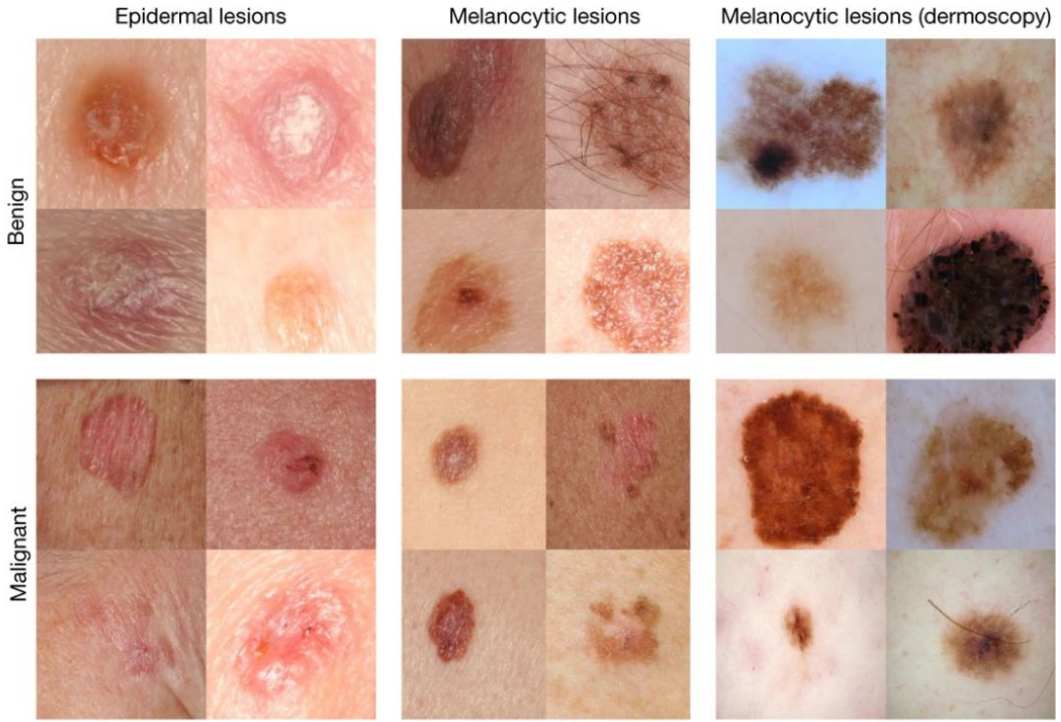
https://learnopencv.com/understanding-convolutional-neural-networks-cnn/

Features are progressively aggregated at higher convolutional layers



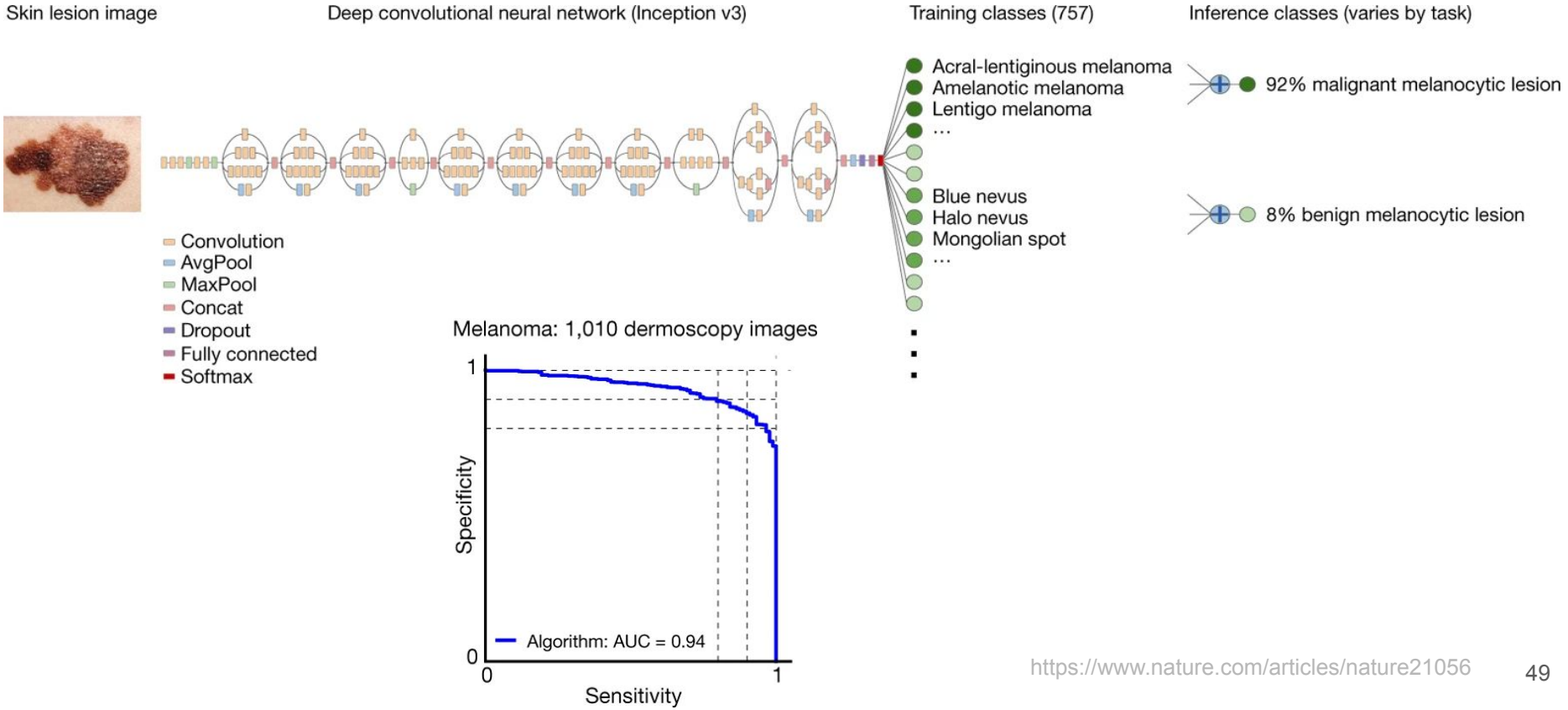
Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

Example: how to predict skin cancers based on imaging?



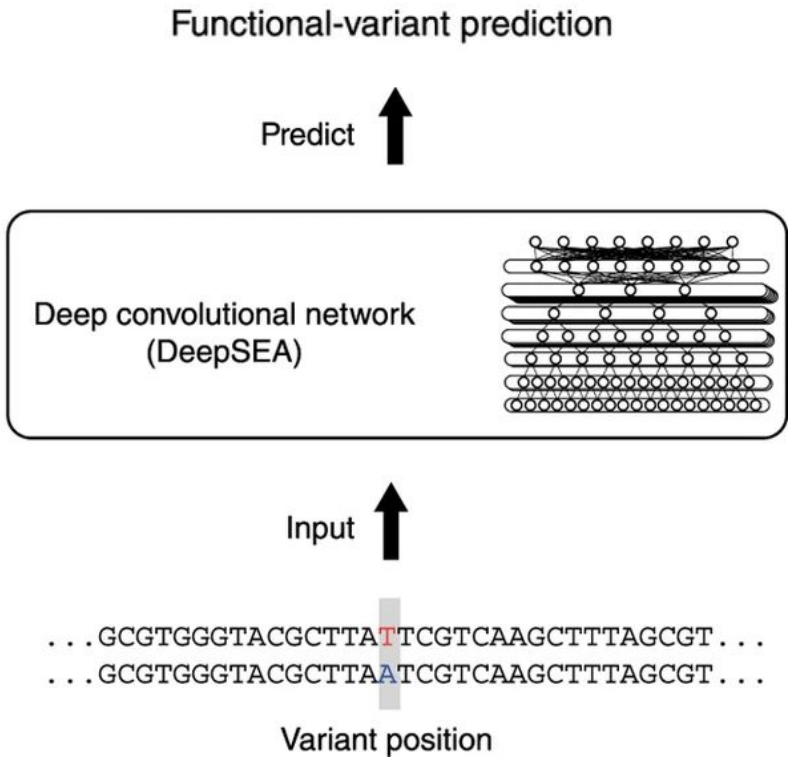
https://www.nature.com/articles/nature21056

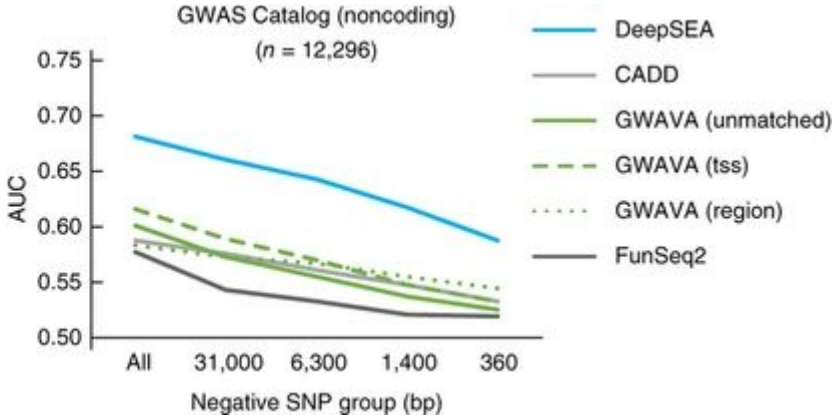
Convolution neural network is able to tell apart cancer v.s. benign cases



https://www.nature.com/articles/nature21056

Example: sequence-based prediction of disease-causing variants





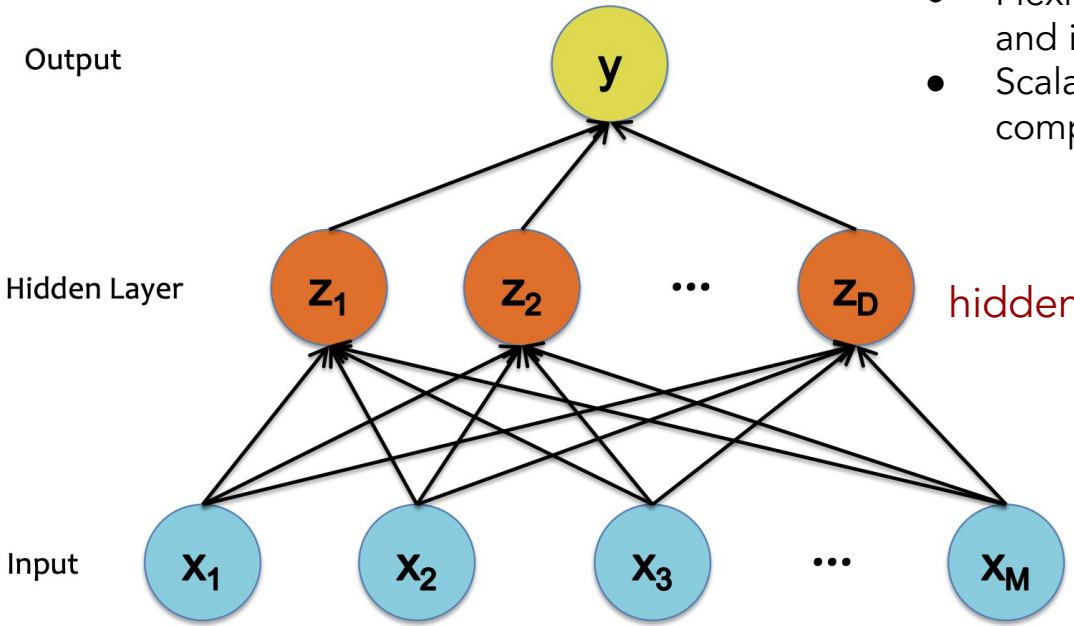
https://www.nature.com/articles/nmeth.3547

Question: when to use neural networks instead of linear models?

A. Predicting gene expression when multiple transcription factors interact nonlinearly.
B. Predicting a phenotype when a simple linear relationship already fits the data well.
C. Classifying cell types from single-cell gene-expression patterns that linear models cannot capture well.
D. Analyzing a small dataset where model interpretability is the primary goal.

Recap

Multilayer Perceptron (MLP)



hidden node

- Nonlinearity: Learns complex, nonlinear relationships among features.
- Flexibility: Captures a wide range of patterns and interactions.
- Scalability: Can benefit from large datasets with complex patterns.